

[Answer any **five** from the following questions. When answering a question, please answer all the subsections of it at once]

1. a) What is the main advantage of the microkernel approach to operating system design? How do user programs and system services interact in micro-kernel architecture? What are the disadvantages of using the microkernel approach? 4
- b) What do you mean by message passing and shared memory approach for inter process communication? 2
- c) What do you mean by context switch and swapping? How they are different? 4
2. a) What resources are used when a thread is created? How do they differ from those used when a process is created? 2
- b) Give a comparative analysis between multi-threading and multi-process programming. 3
- c) Draw the queuing diagram of process scheduling. 3
- d) What does thread provide, concurrency or parallelism? Justify your answer. 2
3. a) Consider the following **Sleeping TA problem** and provide its solution using semaphore. 6

A university computer science department has a teaching assistant (TA) who helps undergraduate students with their programming assignments during regular office hours. The TA's office is rather small and has room for only one desk with a chair and computer. There are three chairs in the hallway outside the office where students can sit and wait if the TA is currently helping another student. When there are no students who need help during office hours, the TA sits at the desk and takes a nap. If a student arrives during office hours and finds the TA sleeping, the student must awaken the TA to ask for help. If a student arrives and finds the TA currently helping another student, the student sits on one of the chairs in the hallway and waits. If no chairs are available, the student will come back at a later time.
- b) What is race condition? Determine whether race condition occur in the following example. 4

```

int          count = 10;

Thread_1(...)      Thread_2(...)
{
    // do something
    count++;
}
                {
    // do something
    count--;
}

```
4. a) What is starvation in operating system? Describe a scheduling algorithm with example which can cause starvation. 2
- b) What are the differences between I/O-bound programs and CPU-bound programs? Why is it important for the scheduler to distinguish between them? 2
- c) Five batch jobs, A through E, arrive at a computer center at almost the same time. The processes are assumed to have arrived in the order A, B, C, D, E. They have estimated running times of 10, 6, 2, 6

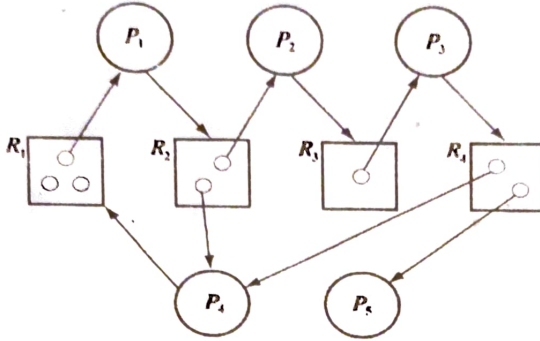
4, and 8 minutes. Their (externally determined) priorities are 3, 5, 2, 1, and 4, respectively, with 5 being the highest priority.

(i) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority and RR (quantum = 2).

(ii) What is the average turnaround time for each of these scheduling algorithms?

(iii) What is the average waiting time for each of these scheduling algorithms?

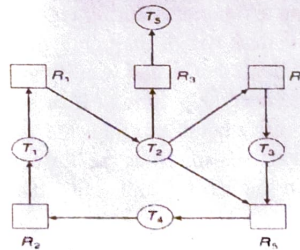
5. a) Consider the following snapshot of a system



Answer the following questions.

- Determine the allocation, need and available matrix from the above figure.
- Is the system in a safe state using the banker's algorithm? Determine the safe sequence if any.

- Discuss how you can detect deadlock in a single instance and multiple instance of resource type. 2
- Consider the following resource allocation graph. Draw the wait for graph and determine whether it is deadlock state or not. 2



- Explain the difference between internal and external fragmentation. 2
- Consider a logical address space of 2,048 pages with a 4-KB page size, mapped onto a physical memory of 512 frames. 6
 - How many bits are required in the logical address?
 - How many bits are required in the physical address?
- The main memory access time is 100ns, and the cache lookup time is 50ns. Assuming a cache hit rate of 90%, what is the average time to read a location from main memory? 2
- Describe the differences between logical and physical addresses. 2
- Consider the following memory partitions (in order): 4

10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB.

(i) How would the first-fit, best-fit, and worst-fit algorithms place processes of size 12 MB, 10MB and 9 MB (in order)?

, Rank the algorithms in terms of how efficiently they use memory.

4

Consider the following page reference string:

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms?

- (i) LRU replacement
- (ii) FIFO replacement

GOOD LUCK!

(Please return this question paper with the answer script)

[Answer any FIVE from the following SEVEN questions]

1. a) Explain how peoples' psychology are linked with organizational behavior. 5
b) Understanding about organizational behavior is critical for all including IT graduates – do you agree? Justify your position. 5
2. a) What is organizational behavior (OB)? What are the major behavioral science disciplines that contribute to the OB field? 5
b) Explain the challenges and opportunities for managers in using organization behavior concepts. 5
3. a) Consider you are appointed as an IT manager in an organization. What roles are desired from you being at the managerial level? Provide at least one example of each role. 5
b) As the manager, consider you are supervising a team of eight people in the IT team. Describe with example the components that would shape the attitude of your subordinates towards you. 5
4. a) What is self-efficacy theory? "Goal setting theory and self-efficacy theory don't compete; they complement each other."- Explain. What are the ways of increasing self-efficacy? 5
b) Two managers are talking about how they get the best out of their employees:
Sazid: "I tell my employees that times are tough and there is no way of telling when the guys in head office might try to downsize. Let me tell you, a bit of fear keeps them focused on the task at hand."
Saikat "I tell them that head office is making them more responsible for deciding how to do their work. The increased sense of ownership they have makes productivity go way up" 5
 - i. How would these managers be best characterized by Douglas McGregor?
 - ii. Differentiate between the assumptions taken by Sazid and Saikat.
5. a) Differentiate between a leader and a manager with examples. 5
b) What is perception? Discuss the link between perception and decision-making with an example. 5
6. a) What is departmentalization? Discuss in how many ways departmentalization can done. Use examples for each way. 5
b) What is discrimination in workplace? Discuss the forms of discrimination with at least one example for each? 5
7. Elaborate the following concepts with examples: (5 x 2 marks) 10
(i) Diversity, (ii) Equity, (iii) Inclusion, (iv) Work Specialization, and (v) Prejudice

Answer any five (5) of the following questions. The weight of each question is mentioned at the right side.

When answering a question, please answer all the subsections of it at once.

1. Consider the sailor database of Figure 1, where the primary keys are underlined.

Sailors (sid, sname, rating, age)
Boats (bid, bname, color)
Reserves (sid, bid, day)

Figure 1. Sailor database.

Give an expression in SQL for each of the following queries.

- a) Find the sailors with a rating above 7. 2
 - b) Find the names of sailors who have reserved all boats. 3
 - c) Find the names of sailors who have reserved a red and a green boat. 2
 - d) Find the names of sailors who have reserved a red or a green boat. 3
2. a) Consider a relation $R(A, B, C, D, E)$, with FDs $A \rightarrow B, CD \rightarrow E$ 4
Is this relation in BCNF? If your answer is yes, explain why it is. If your answer is no, decompose the relation into BCNF, showing your decomposition steps.
- b) What is dependency preserving lossless decomposition? Explain. 3
 - c) Find the canonical cover of the following: 3
- $$R = (A, B, C)$$

$$F = \{ A \rightarrow BC$$

$$B \rightarrow C$$

$$A \rightarrow B$$

$$AB \rightarrow C \}$$
- a) What are the limitations of a file-based system? 3
 - b) What is a DDL? What is a DML? Briefly explain them and give one SQL example of each. 3
 - c) What is a key? Should a key contain only a single value? If yes, explain why. If not, give an example 2
 - d) What are the types of binary relationship sets? Explain them. 2
4. a) Consider the following Information file. 6

ID	Name	City Code
2	M	DHK
3	R	CTG
5	D	RAJ
7	H	SYL
11	A	DHK
17	E	CM

19	U	HG
23	Q	KH
25	A	DHK
29	H	CTG
31	K	RAJ
33	S	SYL

Use the hash function $h(ID) = ID \bmod 8$. Give the

- Static Hash organization for Information file using 8 buckets with each holding a maximum of 2 records.
 - Extendable Hash structure where each bucket can hold a maximum of 2 records.
- b) Assume B+ trees of order $n = 3$ (n is the maximum number of pointers in a node). 2+2

- Draw a tree of height 2 with the following keys: A, C, D, F (lexicographically ordered).
- Show how the insertion of two new keys, followed by their deletions, could change this 2-level tree into a 3-level tree with exactly the same keys (i.e., A, C, D, F). Give an example of such keys, and draw the 3-level tree so generated.

5. Suppose we have our *DhakaSocial* environment a notion of user mailboxes: a "private" inbox for messages (analogous to email) and a public "wall" where others can post (and where messages are visible to all friends). Each message gets a message ID and a visibility level (private or public). Each word is given an ID, and may occur multiple times in each message. We need to keep track of the following attributes:

$R(\text{userID}, \text{msgID}, \text{visibility}, \text{wordID}, \text{wordText}, \text{wordPosition})$

You are given the following FDs:

$\text{userID}, \text{msgID} \rightarrow \text{visibility};$ $\text{wordID} \rightarrow \text{wordText}$
 $\text{wordText} \rightarrow \text{wordID};$ $\text{msgID}, \text{wordPosition} \rightarrow \text{wordID}$
 $\text{msgID} \rightarrow \text{userID};$ $\text{msgID}, \text{wordID} \rightarrow \text{visibility};$
 $\text{msgID}, \text{wordPosition} \rightarrow \text{wordText}$

- Is the FD $\text{wordID}, \text{msgID}, \text{visibility} \rightarrow \text{userID}$ in the closure of the functional dependencies? 2
 - Provide a relational schema in 3NF for the domain, using relational schema notation, indicating keys with underlines under the attributes. 5
 - Is your decomposition lossless? Dependency preserving? 3
6. Consider the following relations where the primary keys are underlined. 10

Doctor(NID, FirstName, LastName, Specialty, YearsOfExperience, PhoneNum)
Patient(NID, FirstName, LastName, Address, DOB, PrimaryDoctor_NID)
Medicine(TradeName, UnitPrice, GenericFlag)
Prescription(Id, Date, Doctor_NID, Patient_NID)
Prescription_Medicine(Prescription Id, TradeName, NumOfUnits)

Write down the relational algebra expressions for the following queries:

- List the trade name of generic medicine with a unit price less than BDT 50.
- List the first and last names of patients whose primary doctor is named 'Deen Islam'.
- List the first and last names of doctors who are not primary doctors to any patient.

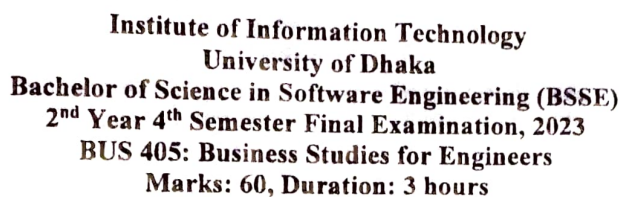
- d) For medicines written in more than 20 prescriptions, report the trade name and the total number of units prescribed.
- e) List the ~~SSN~~ ^{NID} of patients who have 'Aspirin' and 'Vitamin' trade names in one prescription.

7. Consider the following relational table:

Professor (
 professor name, professor id,
 professor office id, student id,
 student name, student office id,
 student designated refrigerator id, refrigerator owner id,
 refrigerator id, refrigerator size, secretary name,
 secretary id, secretary office)

Suppose the data has the following properties:

- A. Professors and secretaries have individual offices, students share offices.
 - B. Students can work for multiple professors.
 - C. Refrigerators are owned by one professor.
 - D. Professors can own multiple refrigerators.
 - E. Students can only use one refrigerator.
 - F. The refrigerator the student uses must be owned by one of the professors they work for.
 - G. Secretaries can work for multiple professors.
 - H. Professors only have a single secretary.
- a) Put this table into 3rd normal form by writing out the decomposed tables; designate keys in your tables by underlining them. Designate foreign keys by drawing an arrow from a foreign key to the primary key it refers to. Note that some of the properties listed above may not be enforced (i.e., guaranteed to be true) by a 3NF decomposition. 6
- b) Which of the eight properties (A-H) of the data are enforced (i.e., guaranteed to be true) by the 3NF decomposition and primary and foreign keys you gave above? 2
- c) What could a database administrator do to make sure the properties not explicitly enforced by your schema are enforced by the database? 2



Respect

1. a) What is Monetary Unit assumption and Economic Entity assumption in accounting? 2
- b) On March 1, 2024 Mr. Joy started Joy Trading Company by investing Tk. 12,000 cash in the business. Following are the assets and liabilities of the company at March 31, 2024 and the revenues and expenses for the month of March.

Cash	Tk. 10,150	Service Revenue	Tk. 6,500
A/R	2,800	Ad expense	500
Supplies	2,000	Rent expense	1,600
Equipment	10,000	Gasoline expense	200
Notes Payable	9,000	Utilities expense	150
A/P	1,200		

Mr. Joy made no additional investment in March but withdraw Tk. 1,300 in cash for personal use during the month. Prepare an income statement and owner's equity statement for the month of March and a balance sheet at March 31, 2024. 3x2
- c) What is the statement of cash flows? What does it report? 4

2. Ms. Fatema is a licensed dentist. During the first month of the operation of her business, the following events and transactions occurred.

April 1 Invested Tk. 20,000 cash in her business.

 - 1 Hired a secretary-receptionist at a salary of Tk. 700 per week payable monthly.
 - 2 Paid office rent for the month Tk. 1,100.
 - 3 Purchased dental supplies on account from Smile Company Tk. 4,000.
 - 10 Provided dental services and billed insurance companies Tk. 5,100.
 - 11 Received Tk. 1,000 cash advance from a patient, Mr. Tapon, for an implant.
 - 20 Received Tk. 2,100 cash for services completed and delivered to a patient Ms. Jyoti.
 - 30 Paid secretary-receptionist for the month Tk. 2,800.
 - 30 Paid Tk. 2,400 to Smile Company for accounts payable due.
 - a) Journalize the transactions. 6
 - b) Prepare corresponding T accounts. 4
 - c) Prepare a Trial Balance on April 30, 2024. 2

3. a) Compare the characteristics of Programmed and Non-programmed decisions. 2
- b) Write the purposes of planning. Give one example of each of the Long-Term plans, Directional Plans, and Standing Plans. 6
- c) Write about the following decision-making errors and biases: 4

Sunk cost error, Hindsight bias, Self-Serving bias, Framing bias

4. a) Illustrate with an example (use both journal entries and corresponding T accounts) how a company closes all the nominal accounts? 6
- b) Write about the classified balance sheet items. 3
- c) Differentiate between Proprietorship, Partnership, and Corporation. 3
5. a) What is the concept of time value of money? Give example. 3
- b) What do you understand by amortization of loans? Discuss with an example. 4
- c) Write about the marketing management orientations of a company to market its products or services. 5
6. a) Discuss the internal conditions affecting group behavior. 3
- b) What is conflict? Write about different types of conflicts. 4
- c) Among the five conflict management techniques (Avoidance, Accommodation, Forcing, Compromise, and Collaboration), which one would you prefer as a manager? Why? 3
- d) What do you understand by bounded rationality? 2
7. Write short notes on the following topics: 3 × 4
 - i. Holistic Marketing concept
 - ii. Empowering Employees
 - iii. Intuitive decision making
 - iv. Herzberg's Motivation-Hygiene Theory



Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
2nd Year 4th Semester Final Examination, 2023
SE 406: Software Requirements Specifications and Analysis
Marks: 50 # Duration: 2.5 hours



Professionalism

Excellence

Respect

[Answer any 5 of the following questions including the question number 4]

Consider the following inception of an IT project:

An Admission Application Management System (AAMS) for XYZ University

[University of XYZ every year arranges an admission test for its undergraduate programs. The University needs an admission application management system to efficiently conduct the admission process.]

There is a Central Admission Team (CAT). The CAT uploads the admission circular in their admission website. CAT collects the HSC passed students' information from Education Boards (EBs). The information contains – Student's name, Mother's name, HSC and SSC Roll, HSC and SSC Registration Number, HSC and SSC GPA and Grades in individual subjects. (For simplicity) Subjects are Physics, Chemistry, Mathematics and Biology.

Interested applicants apply online at the admission website by filling up a form. Applicants first pay 1500 Taka through Mobile Financial Services (MFS). If MFS confirms the payment against a mobile number, applicants can provide their HSC registration number, present address and a recent photograph in the form and their other information is auto filled from the EB data including the mobile number from which payment is made. After providing all the information, the admission system sends an OTP of 6 numeric digits to the assigned mobile number. Applicants need to provide the OTP to the system for confirmation. If the application is successful, a unique serial number is generated and is shown in the website and the same message is sent to the mobile number.

Admission test is held in eight divisional headquarters – Dhaka, Chattogram, Rajshahi, Khulna, Mymensingh, Sylhet, Rangpur and Barishal. Each divisional headquarters has an Admission Test Coordinator (ATC) whose Name, email address and Division are stored. Based on the present address, applicants are allocated in their respective divisions. CAT provides the list of assigned students to ATCs. ATCs arrange exam centers in each divisional headquarters, and assign Center Chiefs (CCs) for each of the exam centers. ATCs upload the center name, address and CCs names and email addresses.

The seats are allocated according to a newly assigned randomly generated Exam Roll numbers to each applicant. Two days before the admission test, the exam rolls are assigned and an admit card is generated. The admit card contains the applicant's name, mother's name, serial and exam roll numbers, photo and the exam center name. The admit cards are downloadable and applicants need to login in the admission website using their serial number. After providing the serial number, an OTP will be sent to the mobile phone. If the OTP is matched, applicants can enter into the website and download the admit card. According to the admit cards, a list of attendance sheet is generated and that is emailed to CCs from CAT. [Login of CAT, ATC and CC do not need to consider.]

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|----|--|----|
| 1. | Draw the use case diagrams for the AAMS software project. | 10 |
| 2. | Draw the activity diagrams of the AAMS project | 10 |
| 3. | Draw ER diagram with proper data modeling approach for the AAMS project. | 10 |
| 4. | Identify the classes with analysis and draw Class Responsibility Collaboration (CRC) modeling diagram. | 10 |
| 5. | Draw the State Transition Diagrams for the AAMS software project. | 10 |
| 6. | Draw the Sequence Diagram for the AAMS project. | 10 |
| 7. | Draw the Data Flow Diagram of the AAMS project. | 10 |

Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
2nd Year 4th Semester Final Examination, 2023
CSE 411: Information Security
Marks: 50 # Duration: 2.5 hours



Professionalism

Excellence

Respect

*Answer any five (5) of the following questions. The weight of each question is mentioned at the right side.
When answering a question, please answer all the subsections of it at once.*

1. a) 'Uses of cryptography in system development is sufficient to make the system secure' – Do you agree with the statement? Justify your answer with reasonable arguments. 4
- b) What is meant by security policy and security mechanism? Explain briefly with a suitable example. 4
- c) Briefly explain the *mutual authentication* and *challenge-response* authentication scheme with a suitable example. 2

2. a) What is meant by block cipher modes of operation? Explain the encryption and decryption process of the *Cipher Block Chaining (CBC) mode of operation* with a suitable diagram. What security is ensured by the CBC mode of operation over the Electronic Code Block (ECB) mode of operation? 1+4+1
- b) Suppose that an intruder has an encrypted message as well as the decrypted version of that message. Can the intruder mount a cipher-text only attack, a known-plaintext, or a chosen-plaintext attack? 4

3. a) Comments on the key size, plaintext block size, number of rounds, round key size, and expanded key size in standard AES (128-bit AES) algorithm. What basic functions are used in AES? 3+1
- b) What is an exhaustive key search? What property is required of an encryption function to ensure that such a search is infeasible? 2
- c) What security services are provided by Digital Signature? Briefly explain the RSA digital signature generation and verification process with a suitable diagram. 1+3

4. a) Explain the key generation algorithm for RSA. Perform encryption and decryption using the RSA algorithm for the following:

$$p = 5; q = 11, e = 3; M = 9$$
3+3
- b) What do you mean by a Social Engineering Attack? Explain at least one attack and prevention mechanism. 4

5. a) 'People is the weakest link in any Information System' – Do you agree with the statement? Justify your answer with reasonable arguments. 4
- b) Briefly explain multi-factor authentication with a suitable example. 2
- c) Briefly explain the four security controls to protect information assets. 4

6. a) Define a collision and explain how to deal with collision attacks in cryptographic hash functions. 4

b) Draw the complete block diagram of a message digest generation using SHA-512. Also, draw the diagram of processing for a single 1024-bit block of a message. 3+3

7. a) Write short notes on the following tools: 6

(i) Wireshark

(ii) Nmap

(iii) Phishing Attack

b) Explain the man-in-the-middle attack in Diffie-Hellman key exchange protocol. 4